

Sampling People, Networks and Records

Week 2 Quiz: Problem Set

Please input your answers for the following problems via the Coursera course site:
<https://www.coursera.org/learn/sampling-methods>.

For the following sample values of salaries from a simple random sample of $n = 20$ faculty selected from $N = 150$, compute the specified statistics.

103, 55, 61, 77, 58, 60, 57, 64, 57, 79, 52, 39, 46, 55, 29, 40, 47, 103, 56, 118

- 1) The sum, $\sum_{i=1}^{20} y_i = (103 + 55 + \dots)$.
- 2) The mean, $\bar{y} = \frac{1}{n} \sum_{i=1}^{20} y_i = \frac{1}{20} (103 + 55 + \dots)$.
- 3) The sampling fraction, $f = \frac{n}{N}$.
- 4) The finite population correction, $\left(1 - \frac{n}{N}\right) = (1 - f)$, where f is the sampling fraction computed in 2 c) above.
- 5) The sum of squared deviations, $\sum_{i=1}^{20} (y_i - \bar{y})^2 = (103 - \bar{y})^2 + (55 - \bar{y})^2 + \dots$, where $\bar{y}$ is the mean computed in 2 b) above.
- 6) The sample variance, $s^2 = \left(\frac{1}{n-1}\right) \sum_{i=1}^{20} (y_i - \bar{y})^2 = \left(\frac{1}{20-1}\right) \left((103 - \bar{y})^2 + (55 - \bar{y})^2 + \dots\right)$.
- 7) The sampling variance of the mean, $v(\bar{y}) = (1 - f) \frac{s^2}{n} = \left(1 - \frac{20}{150}\right) \frac{s^2}{20}$, where s^2 is the sample variance computed in 2 f) above.
- 8) The standard error of the mean, $se(\bar{y}) = \sqrt{v(\bar{y})}$, where $v(\dots)$ is computed in 2 g) above.
- 9) The margin of error, $2 \times se(\bar{y})$, where $se(\dots)$ is computed in 2 h) above.
- 10) We know for the population of faculty that S^2 is 735, and we wanted a standard error of 4 (to give a margin of error of $2 \times \$4 = \8), what simple random sample size would be needed?

